Li-Sheng Zhuang 莊立勝

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Education

National Tsing Hua University — Hsinchu, Taiwan
M.S. in Statistics and Data Science, GPA: 4.00/4.30 — 2023–2025

- Thesis: "Nonparametric Mediation Analysis of Non-Markov Illness-Death Model"
- Advisor: Prof. Yen-Tsung Huang and Wen-Han Hwang

National Chung Hsing University — Taichung, Taiwan
B.S. in Applied Mathematics; Double Major in Finance, GPA: 4.10/4.30 — 2018–2023

Research Interests

Nonparametric methods; statistical learning theory; causal mediation analysis; survival analysis.

Experience

Academia Sinica — Taipei, Taiwan
Research Assistant — 2025–Present

- Supervisor: Prof. Yen-Tsung Huang
- Project: Multivariate Mediation Analysis via SIR-Based Single-Index Models

  Developed a two-stage semiparametric framework for estimating causal mediation effects with multivariate mediators. The method combines sliced inverse regression for dimension reduction with two single-index models for estimation.

National Chung Hsing University — Taichung, Taiwan
Teaching Assistant — 2021–2023

- Courses: Calculus and Ordinary Differential Equations

Publications

[1] L.-S. Zhuang, J.-C. Yu, and Y.-T. Huang, "Non-parametric mediation analysis of non-markov illness-death model", *Statistics in Medicine*, 2026.

Scholarships and Awards

- Poster Competition Winner, 2026 ISI–ISM–ISSAS Joint Conference, Academia Sinica — 2026
- Chen Wen-Cheng Statistical Science Scholarship (陳文成統計科學獎學金) — 2025

  Received for outstanding master's thesis research in statistical science; awarded annually to only two master's students.
- Award of Excellence, The 7th Graduate Student Symposium on Statistical Research — 2025
- Honorary Member of The Phi Tau Phi Scholastic Honor Society (中華民國斐陶斐榮譽學會榮譽會員) — 2022

  Awarded for academic excellence, ranking in the top 1% of the College of Science.
- Outstanding Teaching Assistant Award – Calculus — 2021

Conference Talks

- European Meeting of Statisticians, Lugano, Switzerland 2026
  Topic: Nonparametric Mediation Analysis of Non-Markov Illness-Death Model
- South Taiwan Statistics Conference (南區統計研討會), Taipei, Taiwan 2025
  Topic: Nonparametric Mediation Analysis of Non-Markov Illness-Death Model
- South Taiwan Statistics Conference, Kaohsiung, Taiwan 2024
  Topic: Causal Mediation of Semi-competing Risks Using Interventional Counterfactual Process

Posters

- ISI–ISM–ISSAS Joint Conference, Institute of Statistical Science, Academia Sinica, Taipei, Taiwan 2026
  Topic: Nonparametric Mediation Analysis of Non-Markov Illness-Death Model
- The 7th Graduate Student Symposium on Statistical Research, Taichung, Taiwan 2025
  Topic: Nonparametric Mediation Analysis of Non-Markov Illness-Death Model
- Poster Competition, Institute of Statistical Science, Academia Sinica, Taipei, Taiwan 2025
  Topic: Causal Mediation of Semi-competing Risks Using Interventional Counterfactual Process

Selected Projects

- A General Paired t-Test under Non-i.i.d. Assumptions 2024

  Applied the Lyapunov Central Limit Theorem and k-nearest neighbors to extend the classical paired t-test framework.

- Fourier-Based Stock Price Modeling 2024

  Applied the Fourier transform to denoise log returns in stock price data, improving prediction accuracy in a time series forecasting project.

Certifications

- Society of Actuaries Exam P (Probability)

Technical Skills

I have experience using R, Python, and Matlab for data analysis, model development, and statistical computing.